

Question No 1 [10 pts]

(a) Probability (Number 4 appears 0 times)

$$= 1 - \text{Prob}(1, 2 \text{ or } 3 \text{ appears all the 3 times})$$

$$= 1 - \left(\frac{3}{4}\right)^3 = 0.578 \quad \text{--- 2pts}$$

(b) $P(A \cap B) = P(\text{Sum} = 7)$

$$= P(3, 4) + P(4, 3) \quad \text{--- 1pt}$$

$$= 2/16 \quad \text{--- 1pt}$$

(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A) = P(\text{Sum is odd})$$

$$= P(\text{Sum} = 3) + P(\text{Sum} = 5) + P(\text{Sum} = 7)$$

$$= P((1, 2), (2, 1)) + P((1, 4), (2, 3), (3, 2), (4, 1))$$

$$+ P((3, 4), (4, 3))$$

$$= 8 \times 1/16 = \frac{8}{16} = \frac{1}{2} \quad \text{--- 1pt}$$

$$P(B) = P(\text{Sum} = 6) + P(\text{Sum} = 7) + P(\text{Sum} = 8)$$

$$= P((2, 4), (3, 3), (4, 2)) + P((3, 4), (4, 3)) + P((4, 4))$$

$$= 6/16 \quad \text{--- 1pt}$$

$$P(A \cup B) = \frac{8}{16} + \frac{6}{16} - \frac{2}{16} = \frac{12}{16} = \frac{3}{4} \quad \text{--- 1pt}$$

(d) $P(A \cap B) = 2/16$

$$P(A) \times P(B) = \frac{8}{16} \times \frac{6}{16} = \frac{3}{16} \quad \text{--- 1pt}$$

$\Rightarrow P(A \cap B) \neq P(A) \cdot P(B)$, thus A & B are NOT independent. --- 2pt

Question No 2 [15pts]

(a) Lets Calculate the Probability that student reaches in time

$$P(\text{Reaching in time}) = P(RIT|option1) \times P(option1) + P(RIT|option2) \times P(option2) + P(RIT|opt3) \times P(opt3)$$

$$= 0.85 \times 0.5 + 0.5 \times 0.25 + 0.5 \times 0.25$$

$$= 0.425 + 0.125 + 0.125$$

$$P(RIT) = 0.675$$

$$P(\text{Late}) = 0.325 \Rightarrow 32.5\% \text{ chance of being Late.}$$

$$(b) P(\text{option 2} | RIT) = \frac{P(RIT | \text{option 2}) \times P(\text{option 2})}{P(RIT)}$$
$$= \frac{0.5 \times 0.25}{0.675} = \frac{0.125}{0.675} = 0.185$$

$$(c) E[\text{Commute Time}] = E[CT|opt1] \times P(opt1) + E[CT|opt2] \times P(opt2) + E[CT|opt3] \times P(opt3)$$
$$= \frac{40}{2} \times 0.5 + \frac{25}{2} \times 0.25 + \frac{45}{2} \times 0.25$$
$$= 10 + 3.125 + 5.625$$
$$= 18.75 \text{ minutes}$$

Solution Q3

(a) Activity A



$$= 21$$

1.5

without any Restriction

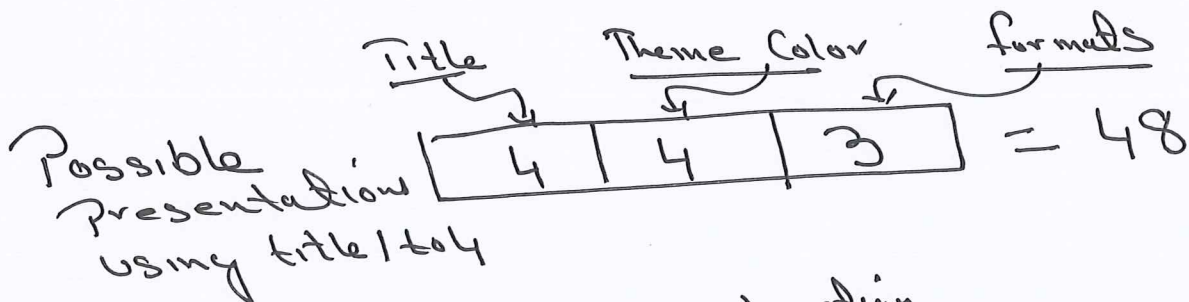
1.5

One of these Combination $\boxed{W_3 | W_1 | W_2} = 1$ NOT allowed

$$\text{Valid Workshop Selections} = 21 - 1 = 20$$

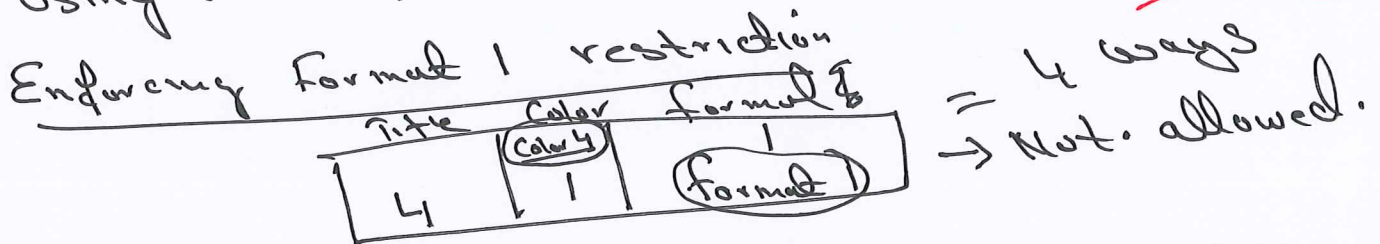
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(b) Activity B Lets find distinct valid presentation for Title 1 to 4 (Title 1, 2, 3, 4)



1 pt

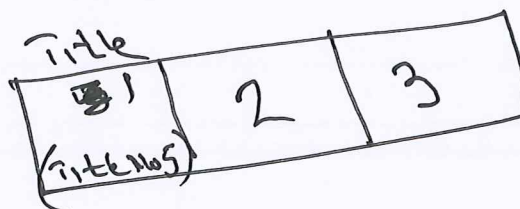
1 pt



$$\Rightarrow \text{Valid (distinct) presentations using title 1, 2, 3, 4} = 48 - 4 = 44$$

1 pt

Now, for Title 5



= 6 valid presentations for title 5.

1 pt

$$\text{Total Valid Presentations} = 44 + 6 = 50$$

1 pt

(c) Activity C

Lets Only Enforce the Vowel Condition and ignore digit condition.

$$\begin{array}{l} 26 \text{ Alphabets} \\ [5 \text{ vowel} + 21 \text{ consonants}] \end{array} + 10 \text{ digits} \quad \textcircled{1} \Rightarrow 31 \text{ ways}$$

$$\textcircled{1.5} \rightarrow \underline{5} \quad \underline{31} \quad \underline{31} \quad \underline{31} \quad \underline{31} = 5 \times 31^4$$

↑
fix Vowel
(for Now)

Vowel position can be Selected in 5 ways, therefore

$$\text{Total permutations w/ ignoring digit condition} = 5 \times 5 \times 31^4$$

Now, Lets Enforce digit condition.

determine ~~cases~~ codes with NO digits (ONLY Consonants)

$$\textcircled{1.5} \rightarrow \underline{5} \quad \underline{21} \quad \underline{21} \quad \underline{21} \quad \underline{21} = 5 \times 21^4$$

Vowel again vowel can be Selected in 5 ways.

$$\Rightarrow \text{Not allowed codes} = 5 \times 5 \times 21^4$$

$$\begin{aligned} \Rightarrow \text{Valid codes} &= 5 \times 5 \times 31^4 - 5 \times 5 \times 21^4 \quad \textcircled{0.5} \\ &= 5 \times 5 (31^4 - 21^4) \\ &= 5 \times 5 (729,040) \quad \textcircled{0.5} \\ &= 18,226,000 \end{aligned}$$

Problem 4 [20pts]

(a) 6-sided die has No 1 on 1 side, No 2 on 2 sides and No 3 on 3 sides

3-sided die

$$\Rightarrow \begin{array}{ll} 1 & \text{w.p. } 1/6 \\ 2 & \text{w.p. } 2/6 \\ 3 & \text{w.p. } 3/6 \end{array}$$

2 pt

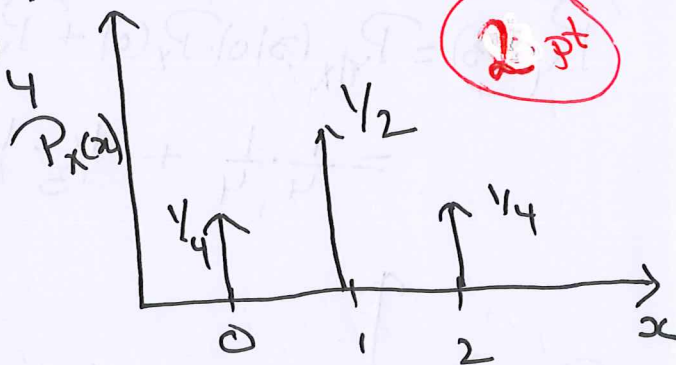
$P(\text{No 3}) = 1/2$, Probability $P(\text{Not getting 3}) = 1/2$.

RV X is No of times 3 appears in 2 Rolls

$$P(X=0) = \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^2 = 1/4$$

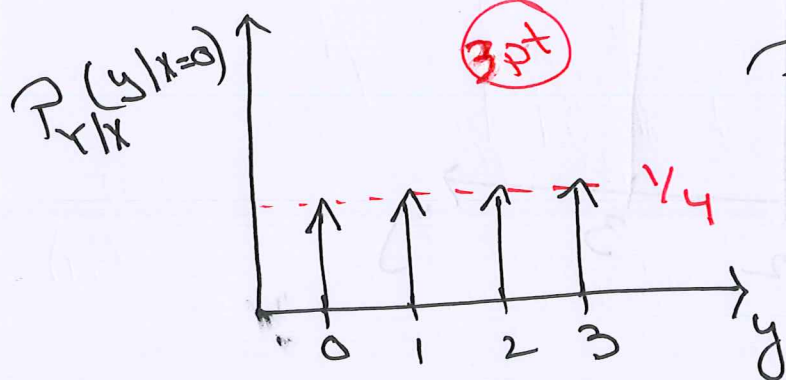
$$P(X=1) = \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^1 \times {}^2C_1 = 1/2$$

$$P(X=2) = \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^0 = 1/4$$



2 pt

(b) RV $Y|X$ is DRV from $x-3$

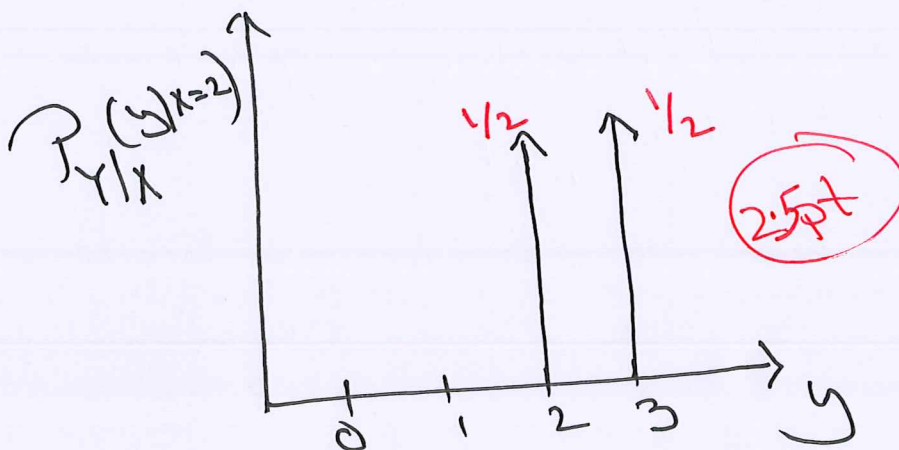


3 pt



2.5 pt

1/3



2.5 pt

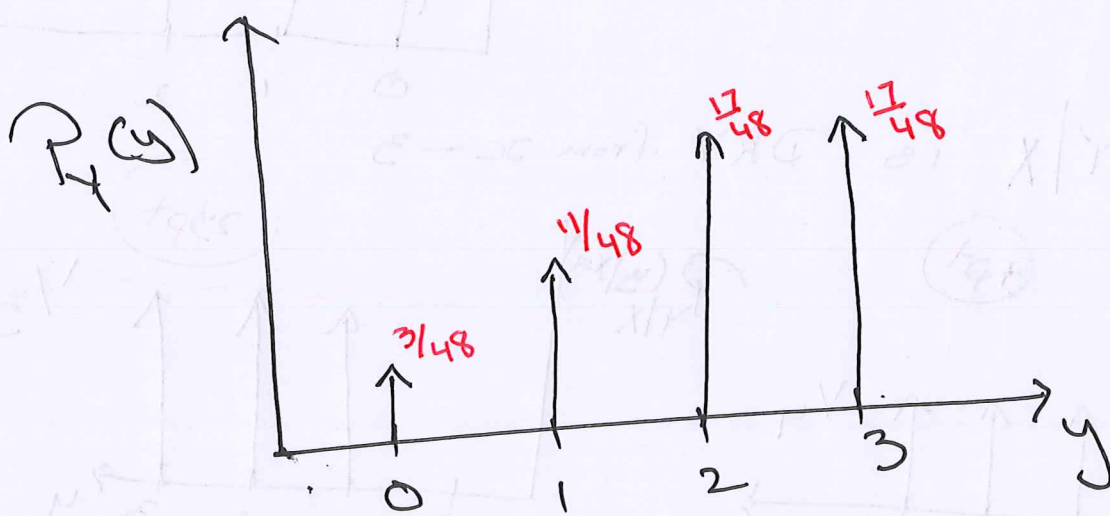
$$(c) P_Y(y) = \sum_x P_{Y|X}(y|x) \cdot P_X(x) \quad \text{1 pt}$$

$$P_Y(0) = P_{Y|X}(0|0) \cdot P_X(0) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} = \frac{3}{48} = 0.0625 \quad \text{1 pt}$$

$$P_Y(1) = P_{Y|X}(1|0) \cdot P_X(0) + P_{Y|X}(1|1) \cdot P_X(1) = \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{2} = \frac{11}{48} = 0.229 \quad \text{1 pt}$$

$$P_Y(2) = P_{Y|X}(2|0) \cdot P_X(0) + P_{Y|X}(2|1) \cdot P_X(1) + P_{Y|X}(2|2) \cdot P_X(2) \\ = \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{17}{48} = 0.354 \quad \text{1 pt}$$

$$P_Y(3) = P_{Y|X}(3|0) \cdot P_X(0) + P_{Y|X}(3|1) \cdot P_X(1) + P_{Y|X}(3|2) \cdot P_X(2) \\ = \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{17}{48} = 0.354 \quad \text{1 pt}$$



4 pt